# Regression and correlation

- Regression is the study of dependency.
- It is used to answer:

(a) Does changing the class size affect the success of student?

(b) Can we predict height of a daughter from the height of mother?

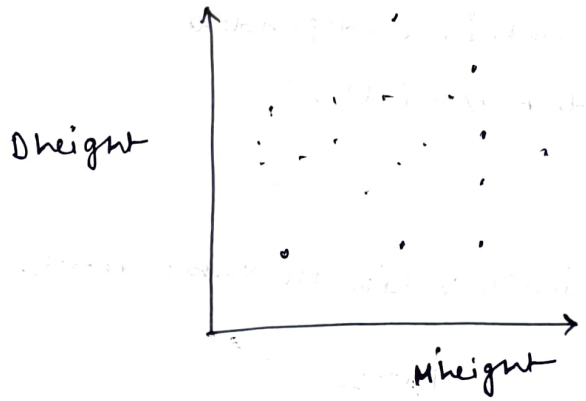
- For this firstly, we need data.
- Then, start with one predicter variable (X) and one response variable (Y)

Data consist of values $(x_i, y_i)$ $i = 1, 2, \ldots, n$ of $(X, Y)$ observed on each of n unit cases.

Example! Inheritance of height

The collection of mother's height (M height) and corresponding daughter's height (D height) - inch

scatter plot

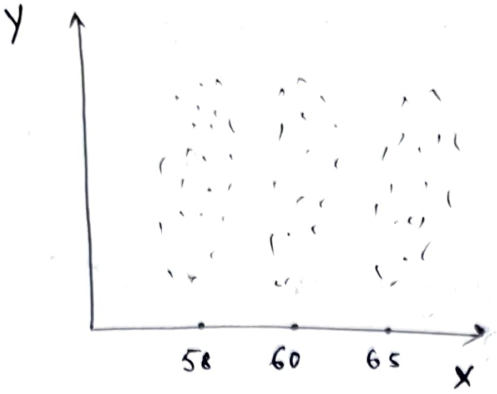



Answer

Do taller mother tend to have taller daughter?
Do shorter mother tend to have shorter daughter?

→ more understandable plot.

Scatter plot showing only pairs where mothers height round to 58, 64, 68 inch




→ Here, we may notice some trends in data.

→ Our interest how the distribution of y changes as x is varied.

→ One import aspect of distribution is mean function and variance function.

## Correlation

A correlation describe a relationship between two variables.

Correlation: is there a relationship between two variables?

Regression: how well a certain independent variable predict dependent variable?

## Type of correlation

Positive: If both the variable varies in same direction.

| Height | 158 | 160 | . . . . | 172 |
|---|---|---|---|---|
| weight | 52 | 55 | 60 | |

## Negative Correlation

| price of a product | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|
| demand | 72 | 120 | 175 | 250 | 260 | |

## Zero correlation

If we don't have any relation between the two variables.

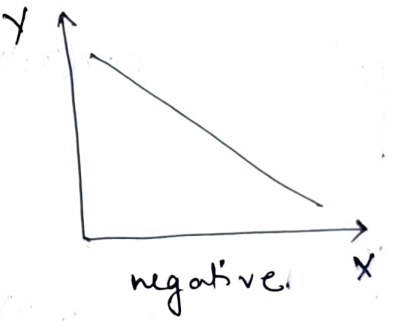
height → intelligence

## Measurement of correlation

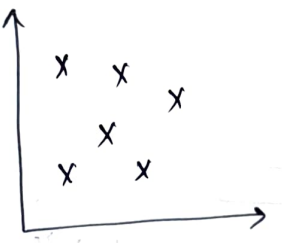
① Scatter plot




positive




negative



Zero correlation

## Variance

Give information on variability of a single variable

$$s_x^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

* $(n-1)$ due to Bessel correction is the adjustment we make when calculating the sample variance instead of population variance.

## Co-variance

$$cov(x,y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

## Calculation of co-variance

1.

| x | y |
|---|---|
| 2 | 20 |
| 3 | 60 |
| 6 | 80 |

$cov(x,y) = 56.67$

2.

| x | y |
|---|---|
| 2 | 30 |
| 4 | 60 |
| 6 | 80 |
| 8 | 100 |

$cov(x,y) = ?$

3.

| x | y |
|---|---|
| 1 | 4 |
| 2 | 5 |
| 3 | 4 |

$cov(x,y) = ?$

## Problem in covariance

- dependent on size of the data's standard deviation
- If large, value is large, if small, then value is small.

| X | Y |
|---|---|
| 101 | 100 |
| 81 | 80 |
| 61 | 60 |
| 51 | 50 |
| 41 | 40 |
| 21 | 20 |
| 1 | 0 |
| 51 | 50 |

covariance = 1166.67

| x | y |
|---|---|
| 54 | 53 |
| 53 | 52 |
| 52 | 51 |
| 51 | 50 |
| 50 | 49 |
| 49 | 48 |
| 48 | 47 |
| 51 | 50 |

Covariance = 4.67

## Solution / Pearson's r (Pearson correlation coefficient)

- Covariance does not really tell us anything
  Solution: standardise this measure
- Pearson's r: standardise the covariance
- Deride the covariance by the multiplied standard deviations of x and y.

$$r_{xy} = \frac{cov(x,y)}{s_x s_y}$$

$$r_{xy} = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\sum (x-\bar{x})^2}\sqrt{\sum (y-\bar{y})^2}}$$

Exercise:

① Find pearson correlation coefficient between advertisement expanses and sales volumn.

Advertisement Exp (in Lakh) : 11, 13, 14, 16, 16, 15, 15, 14, 13, 13

sales volumn (in Lakh) : 50, 50, 55, 60, 65, 65, 65, 60, 60, 50

2. Find the correlation coefficient between age and playing habits of the following students using karl pearson.

| Age | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|
| no of student | 250 | 200 | 150 | 120 | 100 | 80 |
| Regular Player | 200 | 150 | 90 | 48 | 30 | 12 |

Percentage of playing habit

$$= \frac{\text{No. of regular player}}{\text{Total no. of student}} \times 100$$

Pearson's (measure linear relationship between X and Y)

range $-1 \leq r \leq 1$

- $r = 1$ Perfect positive linear correlation
- $r = -1$ Perfect negative linear correlation
- $r = 0$ NO linear correlation.
  → still there may be non-linear one

→ close to (+1) or (−1) strong linear relationship

→ close to (0) weak or no linear relationship

Exercise

Calculate Pearson's r where $y = x^2$ and x ranges from −3 to +3?

$X = [-3, -2, -1, 0, 1, 2, 3]$

Note: There are some correlation coefficient for non-linear relationship

— Spearman's rank correlation coefficient

# Recap one analysis of data

$\bar{x}$ = sample average of $x = \frac{\sum x_i}{n}$

$\bar{y}$ = sample average of $y = \frac{\sum y_i}{n}$

$s_x$ = sample standard deviation

$s_x^2$ = sample variance of $x = \frac{\sum (x_i - \bar{x})^2}{n-1}$

$s_y$ = sample standard deviation

$s_y^2$ = sample variance of $y = \frac{\sum (y_i - \bar{y})^2}{n-1}$

$cov(x,y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$ = sample covariance

$r_{xy} = \frac{cov(x,y)}{s_x s_y}$ = sample correlation.